Chapter 9. Metal–Semiconductor and Semiconductor Heterojunctions

PREVIEW:

1. Determine the energy-band diagram of a metal–semiconductor junction
2. Investigate the electrostatics of the rectifying metal–semiconductor which is known as the Schottky barrier diode.
3. Derive the ideal current–voltage relation of the Schottky barrier diode
4. Discuss differences in the current transport mechanism between the barrier diode and pn junction diode, and discuss differences in turn and switching times.
5. Discuss ohmic contacts, which are low-resistance, nonrectifying metal semiconductor junctions.
6. Investigate the characteristics of a semiconductor heterojunction

Concepts:

Homojunction: The semiconductor material is homogeneous throughout the structure.
Heterojunction: The junction formed by the contact between two different semiconductor materials.
Metal-Semiconductor Junction: Formed between a metal and a semiconductor
Ohmic Contact: A low-resistance metal–semiconductor contact providing conduction in both directions between the metal and semiconductor.
Work function: The minimum energy required to remove an electron

The metal work function: $$e\phi_m = E_0 - \left(E_F\right)_m$$

The semiconductor work function: $$e\phi_s = E_0 - \left(E_F\right)_s = E_0 - E_c + E_c - \left(E_F\right)_s$$
$$= \chi + \left[E_c - \left(E_F\right)_s\right] = \chi + \phi_n$$

9.1 The schottky barrier diode

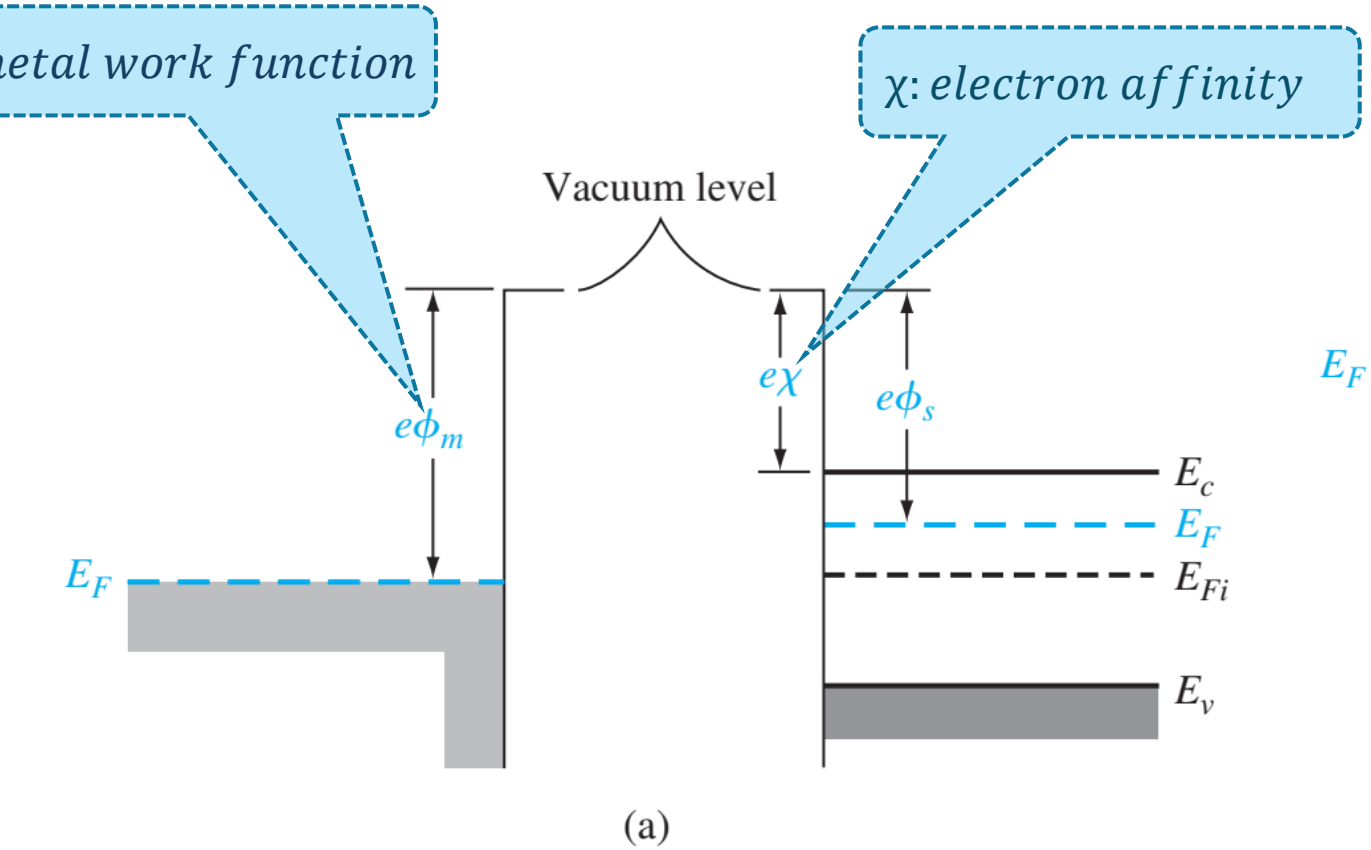
Table 9.1 | Work functions of some elements

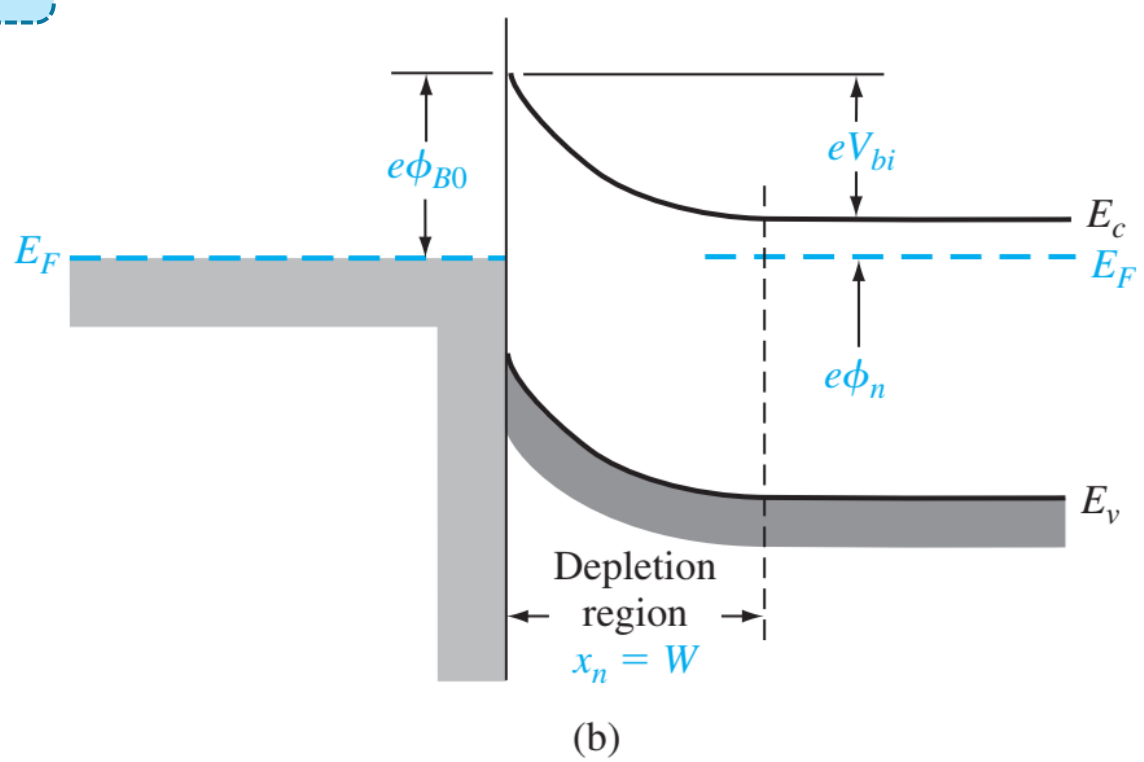
Element	Work function, ϕ_m
Ag, silver	4.26
Al, aluminum	4.28
Au, gold	5.1
Cr, chromium	4.5
Mo, molybdenum	4.6
Ni, nickel	5.15
Pd, palladium	5.12
Pt, platinum	5.65
Ti, titanium	4.33
W, tungsten	4.55

Table 9.2 | Electron affinity of some semiconductors

Element	Electron affinity, χ
Ge, germanium	4.13
Si, silicon	4.01
GaAs, gallium arsenide	4.07
AlAs, aluminum arsenide	3.5

9.1.1 Qualitative Characteristics





(a) Energy-band diagram of a metal and semiconductor before contact;

(b) ideal energy-band diagram of a metal–n-semiconduc$\phi_m > \phi_s$on for

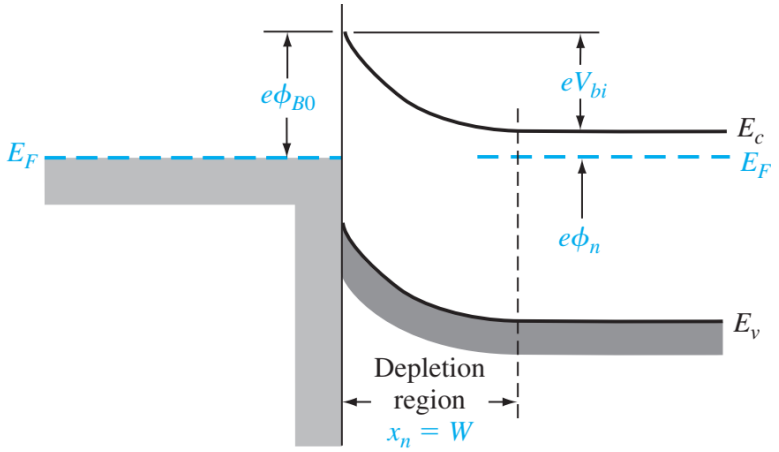
$$\phi_{B0} = (\phi_m - \chi)$$

$$V_{bi} = \phi_{B0} - \phi_n$$

9.1.2 Ideal Junction Properties

Poisson' s equation:

$$\frac{d^2\phi(x)}{dx^2} = \frac{-\rho(x)}{\epsilon_s} = -\frac{d\mathrm{E}(x)}{dx}$$



$$\mathrm{E} = \int \frac{eN_d}{\epsilon_s} dx = \frac{eN_d x}{\epsilon_s} + C_1$$

Boundary condition: $x = x_n, E = 0$

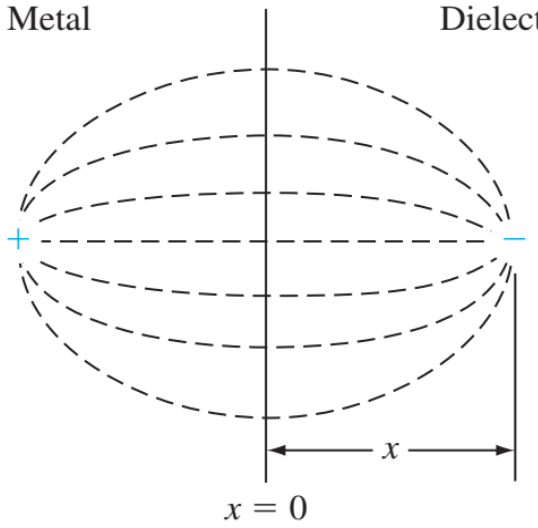
$$C_1 = -\frac{eN_d x_n}{\epsilon_s}$$

$$\boxed{\mathrm{E} = -\frac{eN_d}{\epsilon_s}(x_n - x)}$$

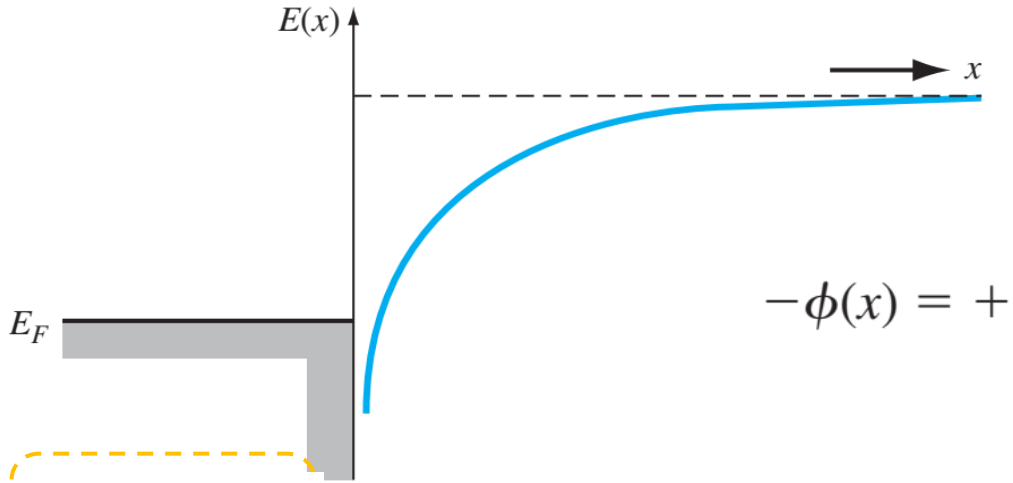
Identical to that of a one-sided pn junction

$$\boxed{W = x_n = \left[\frac{2\epsilon_s(V_{bi} + V_R)}{eN_d}\right]^{1/2}}$$

9.1.3 Nonideal Effects on the Barrier Height (Schottky Barrier Lowering

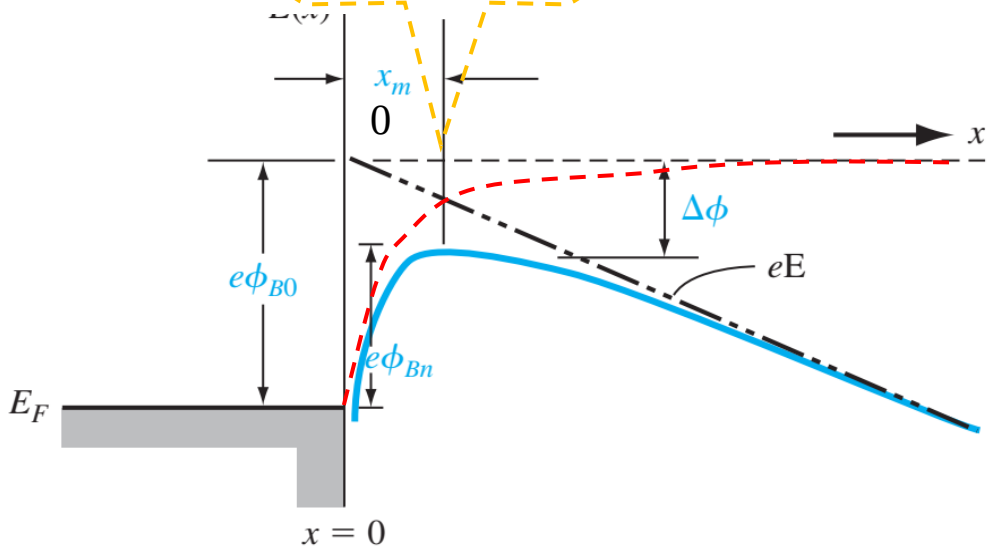


(a)



(b)

$$\frac{d[e\phi(x)]}{dx} = 0$$



(c)

$$F = \frac{-e^2}{4\pi\epsilon_s(2x)^2} = -e\mathrm{E}$$

$$-\phi(x) = +\int_x^\infty \mathrm{E}\,dx' = +\int_x^\infty \frac{e}{4\pi\epsilon_s \cdot 4(x')^2}\,dx' = \frac{-e}{16\pi\epsilon_s x}$$

With an electric field

$$-\phi(x) = \frac{-e}{16\pi\epsilon_s x} - \mathrm{E}x$$

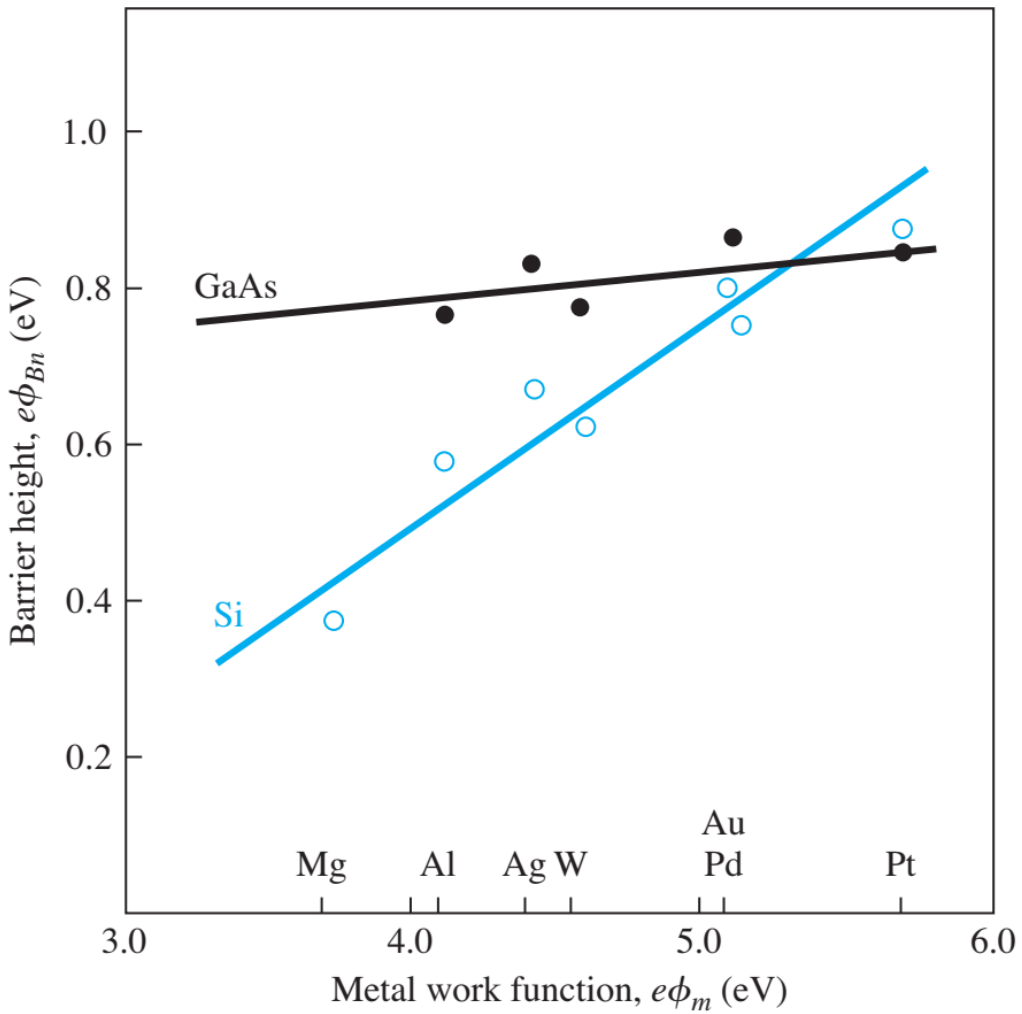
$$x_m = \sqrt{\frac{e}{16\pi\epsilon_s \mathrm{E}}}$$

$$\boxed{\Delta\phi = \sqrt{\frac{e\mathrm{E}}{4\pi\epsilon_s}}}$$

9.1.3 Nonideal Effects on the Barrier Height (Interface States)

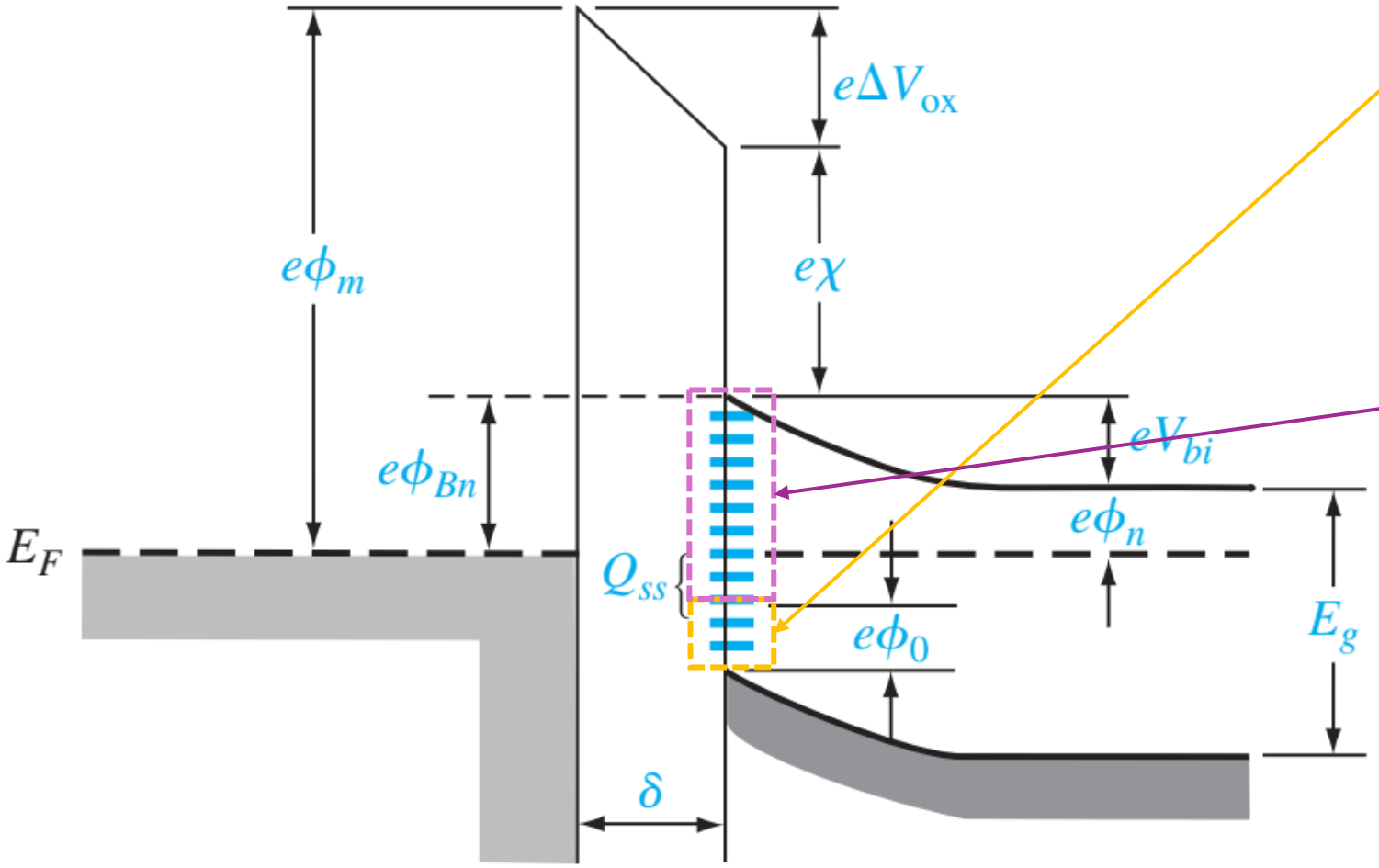
Semiconductor	Metal	$\phi_{Bn}(V)$
n-Ge	Au	0.45
	Al	0.48
	W	0.48
n-Si	Au	0.79
	W	0.67

Semiconductor	Metal	$\phi_{Bn}(V)$
n-GaAs	Au	0.95
	Ag	0.93
	Al	0.80
	W	0.71
	Pt	0.94



Not fit the simple relation given in Equation (9.1)! !

9.1.3 Nonideal Effects on the Barrier Height (Interface States)



Donor states

Contain an electron: Neutral

Not contain an electron: Positively

Acceptor state

Not contain an electron: Neutral

Contain an electron: Negatively

9.1.3 Nonideal Effects on the Barrier Height (Interface States)

$$(E_g - e\phi_0 - e\phi_{Bn}) = \frac{1}{eD_{it}}\sqrt{2e\epsilon_s N_d(\phi_{Bn} - \phi_n)} - \frac{\epsilon_i}{eD_{it}\delta}[\phi_m - (\chi + \phi_{Bn})]$$

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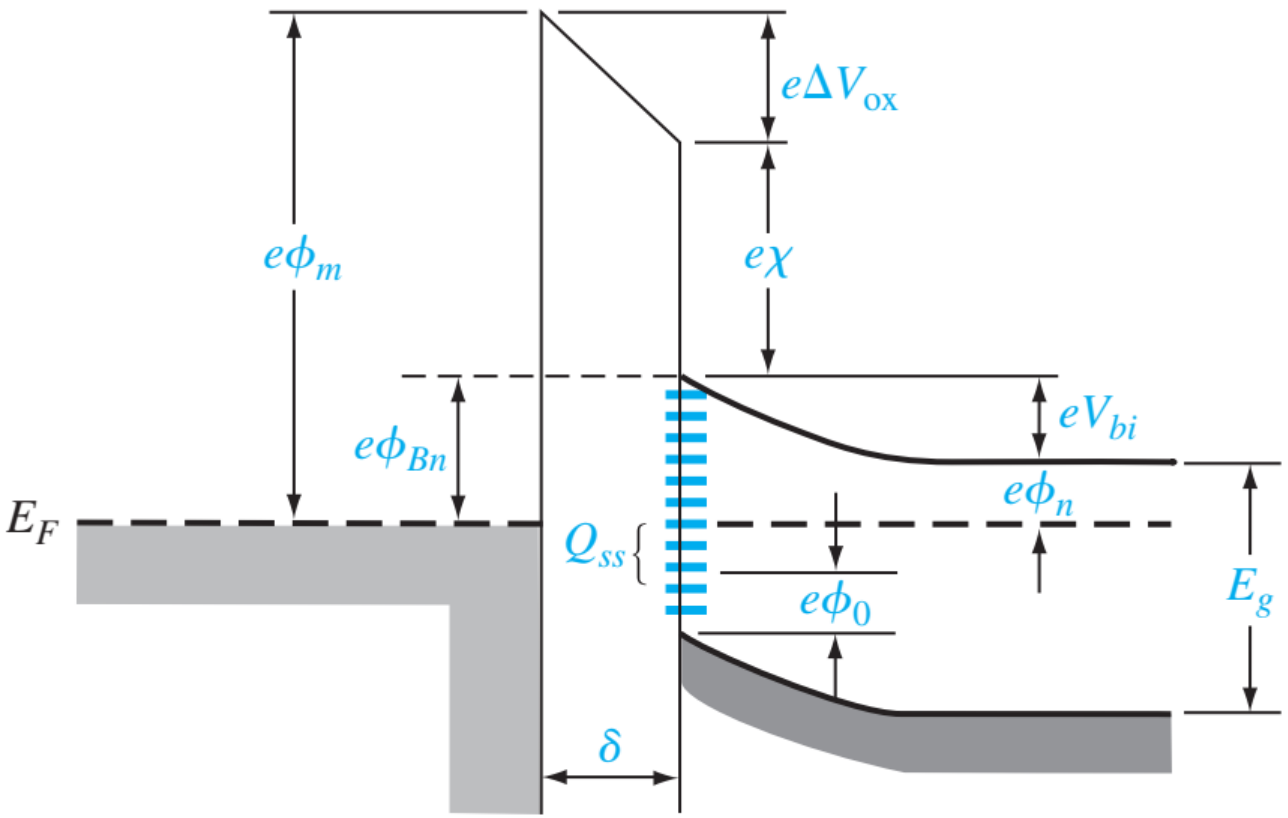
Case 1 Let $D_{it} \to \infty$.

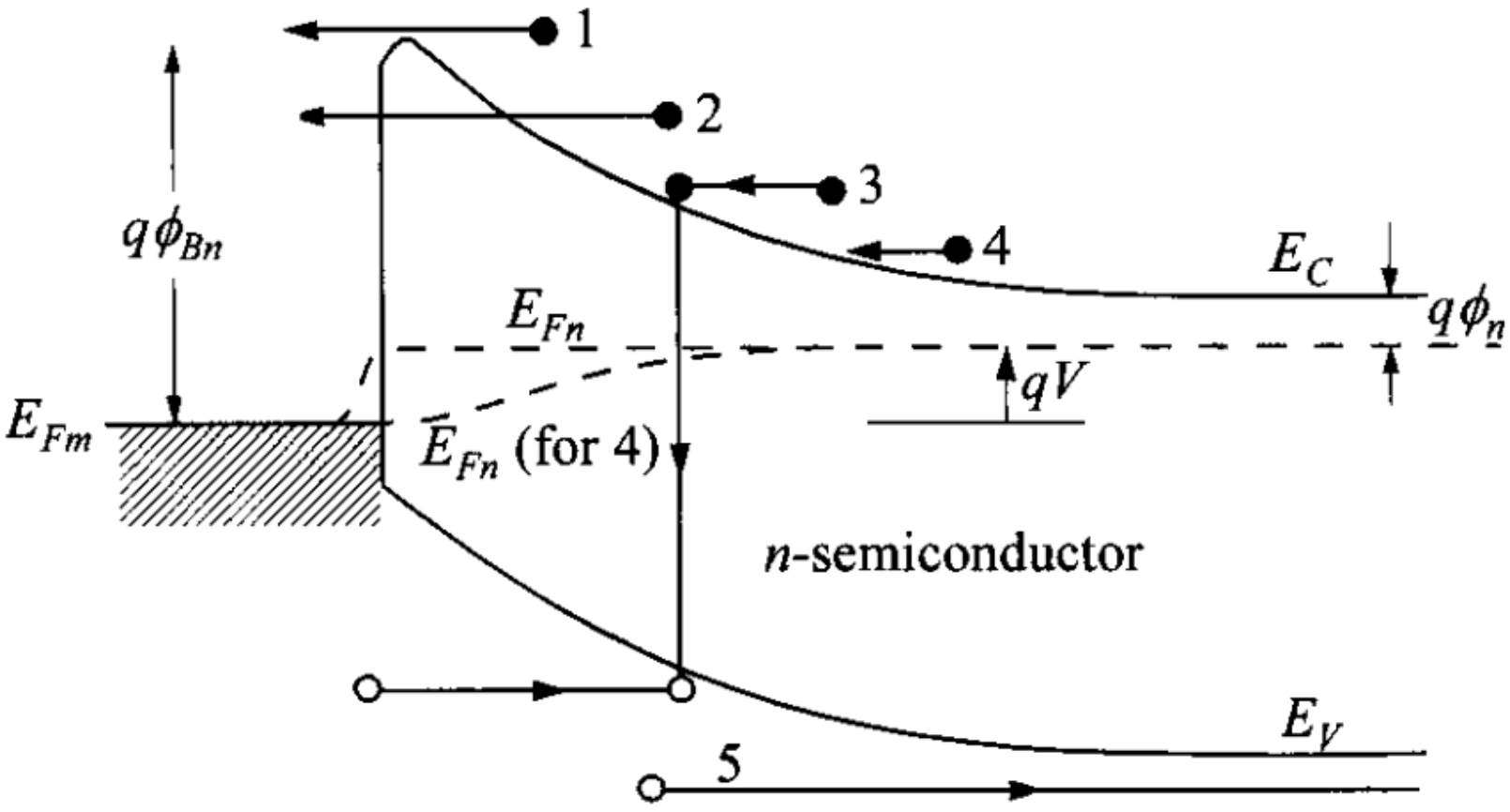
$$\phi_{Bn} = \frac{1}{e}(E_g - e\phi_0)$$

Case 2 Let $D_{it}\delta \to 0$.

$$\phi_{Bn} = (\phi_m - \chi)$$



9.1.4 Current–Voltage Relationship



Five basic transport processes (forward bias.)

(1) Thermionic emission.

(2) Tunneling.

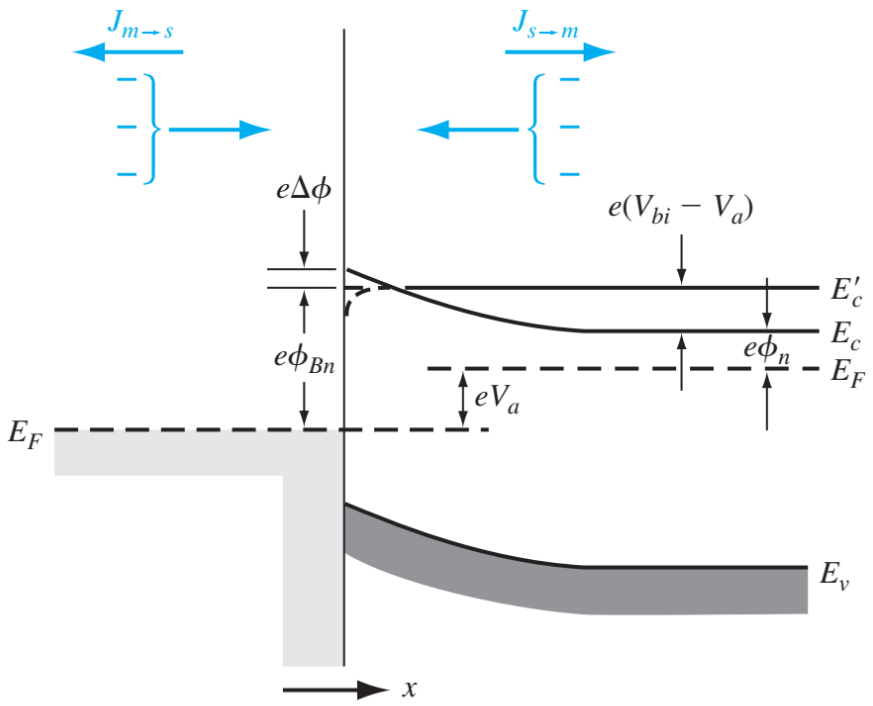
(3) Recombination.

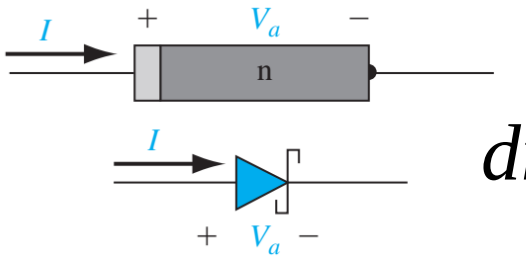
(4) Diffusion of electrons.

(5) Diffusion of holes.

9.1.4 Current–Voltage Relationship

The current transport in a metal–semiconductor junction is due mainly to majority carriers as opposed to minority carriers in a pn junction.





Thermionic emission theory

$$J_{s\to m} = e\int_{E_c'}^{\infty} v_x\,dn$$

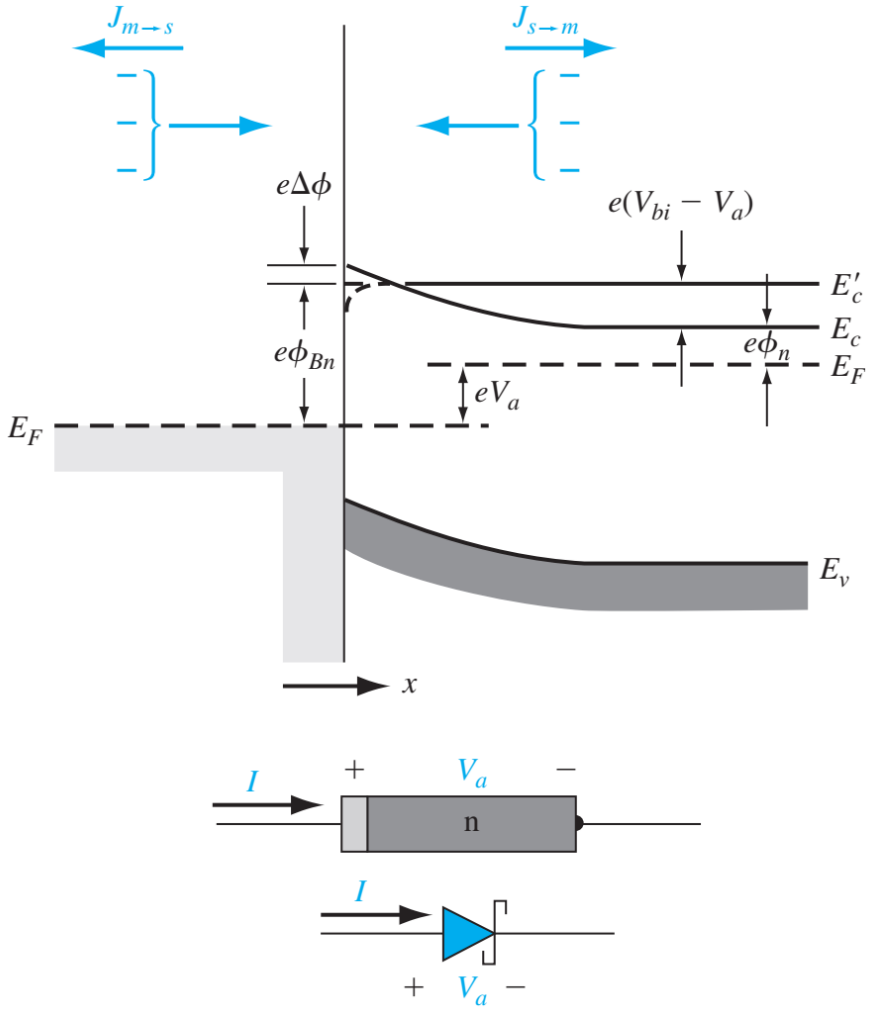
$$E - E_c = \frac{1}{2}m_n^* v^2 \geq e\left(V_{bi} - V_a\right)$$

$$E \geq E_c + e\left(V_{bi} - V_a\right) = E_c'$$

$$dn = g_c\left(E\right) f_F\left(E\right) dE = \frac{4\pi\left(2m_n^*\right)^{3/2}}{h^3}\sqrt{E - E_c}\,\exp\left(-\frac{E - E_F}{kT}\right)dE$$

9.1.4 Current–Voltage Relationship

The current transport in a metal–semiconductor junction is due mainly to majority carriers as opposed to minority carriers in a pn jun



$$\frac{1}{2} m_n^* v^2 = E - E_c$$

$$J = J_{s\to m} - J_{m\to s}$$

$J_{m\to s} = J_{s\to m}(V_a=0)$

$$J = \left[A^* T^2 \exp\left(\frac{-e\phi_{Bn}}{kT}\right)\right]\left[\exp\left(\frac{eV_a}{kT}\right) - 1\right]$$

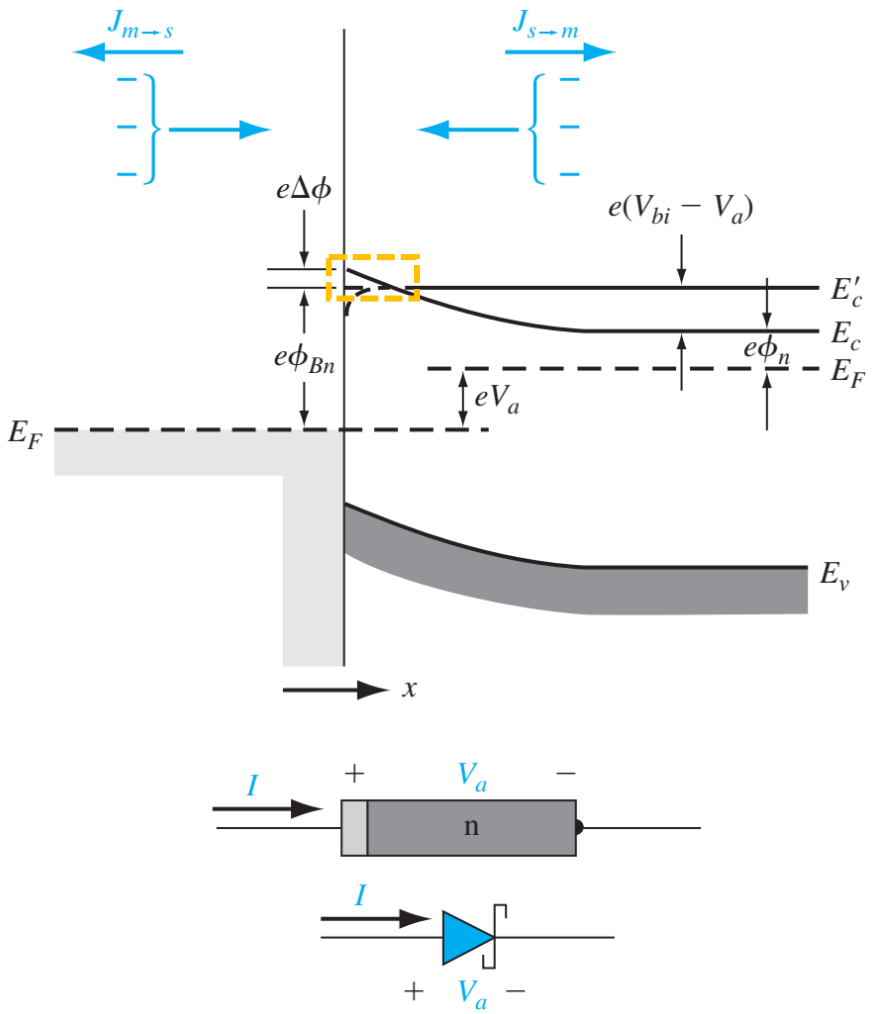
Richardson constant: $A^* \equiv \dfrac{4\pi e m_n^* k^2}{h^3}$

$$\boxed{J = J_{sT}\left[\exp\left(\frac{eV_a}{kT}\right) - 1\right]}$$

$$\boxed{J_{sT} = A^* T^2 \exp\left(\frac{-e\phi_{Bn}}{kT}\right)}$$

9.1.4 Current–Voltage Relationship

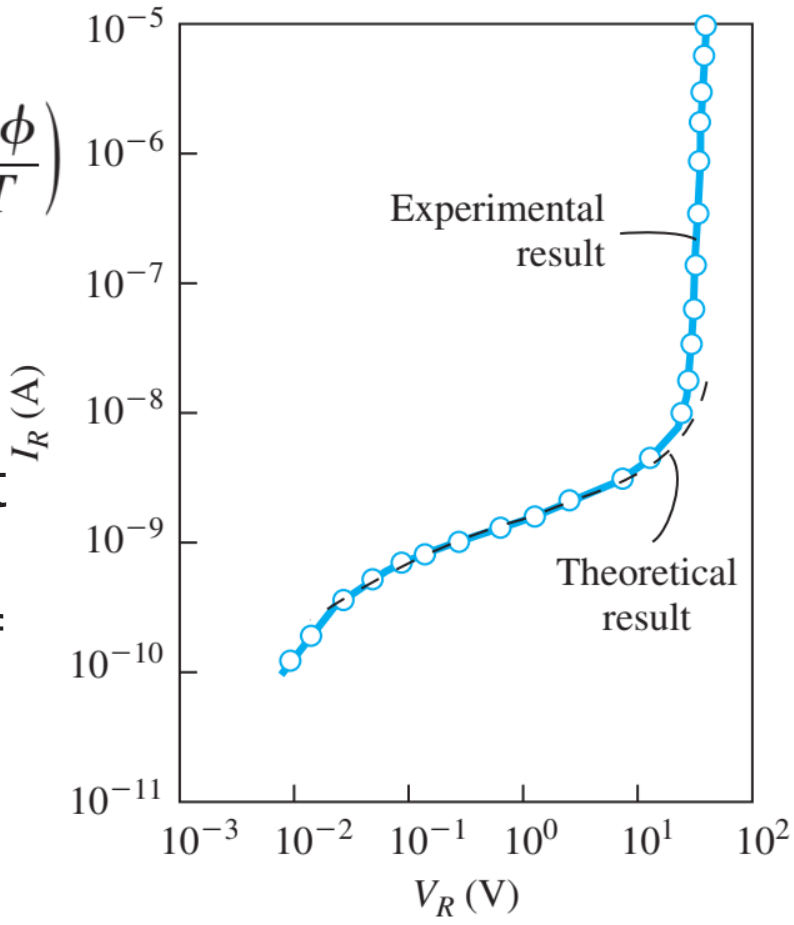
The current transport in a metal–semiconductor junction is due mainly to majority carriers as opposed to minority carriers in a pn junction.



$$\phi_{Bn} = \phi_{B0} - \Delta\phi$$

$$J_{sT} = A^*T^2 \exp\left(\frac{-e\phi_{B0}}{kT}\right) \exp\left(\frac{e\Delta\phi}{kT}\right)$$

The reversebiased current increases with reverse-biased voltage because of the barrier lowering effect.



9.1.5 Comparison of the Schottky Barrier Diode and the pn Junction Diode

Differences (SBD&PNJ)

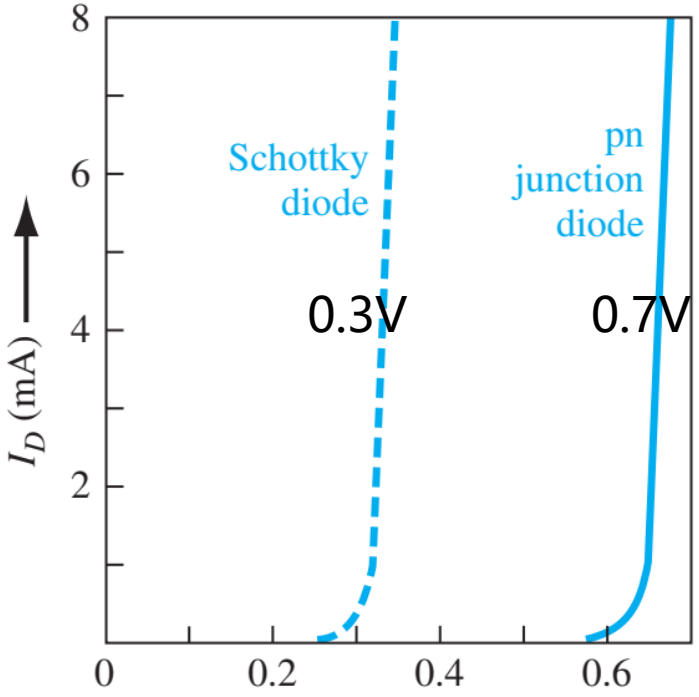
1. The magnitudes of the reverse-saturation current densities

- The current mechanism in the two devices is different.
- The effective turn-on voltage of the Schottky diode is less than that of the pn junction diode.
- The ideal reverse-saturation current density of magnitude larger than that of the ideal PN

7A

$$J_{sT} = A^* T^2 \exp\left(\frac{-e\phi_{Bn}}{kT}\right)$$

$$J_s = \frac{eD_n n_{po}}{L_n} + \frac{eD_p p_{no}}{L_p}$$

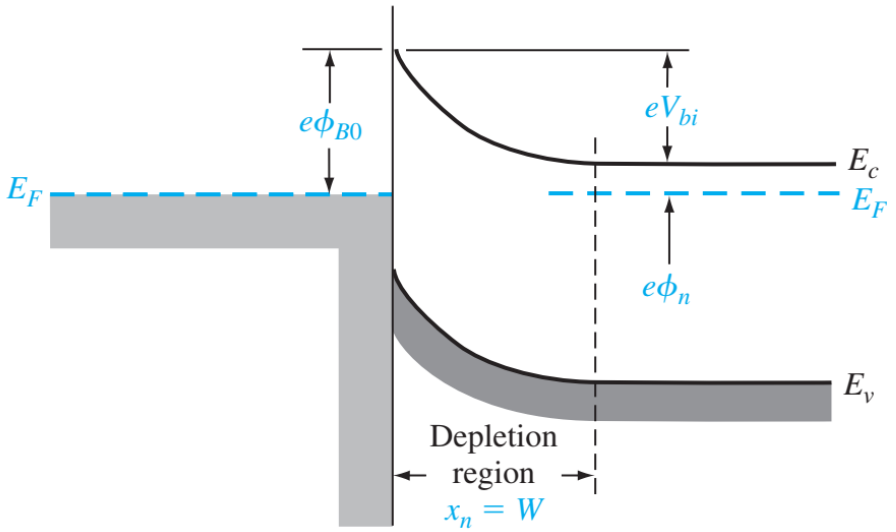


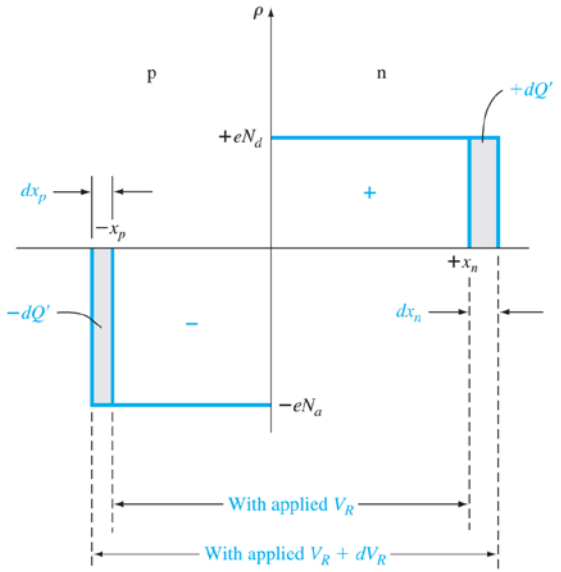
9.1.5 Comparison of the Schottky Barrier Diode and the pn Junction Diode

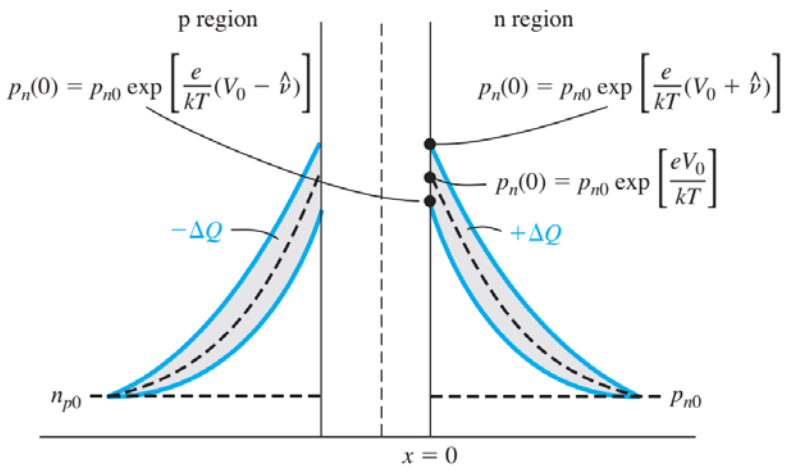
Differences (SBD&PNJ)

1. The switching characteristics (frequency response)

- The current in a SBD is attributed to the injection of majority carriers across a potential barrier.
- Switching a Schottky diode from forward to reverse bias, there is no minority carrier stored charge to remove, as is the case in the pn junction diode. SBD can be used in fast-







9.1.5 Comparison of the Schottky Barrier Diode and the pn Junction Diode

	肖特基势垒二极管	PN结二极管
器件类型及特点	多子器件，载流子无存贮	少子器件，电荷存贮效应
导通电压	0.3~0.5V	0.5~0.7V
最高工作频率	高	低
耐压	低，一般低于100V	高，可达1000V
反向恢复时间	小于10ns	普通二极管500ns~5μs/快速恢复150ns~500ns/超快恢复10ns~150ns
反向漏电流	高（500μA@25°C）	低（5μA@25°C）

9.2 Metal–Semiconductor Ohmic Contacts

Ohmic contacts*:* the current through the ohmic contact is a <u>linear</u> function of applied voltage, and the applied voltage should be very small.

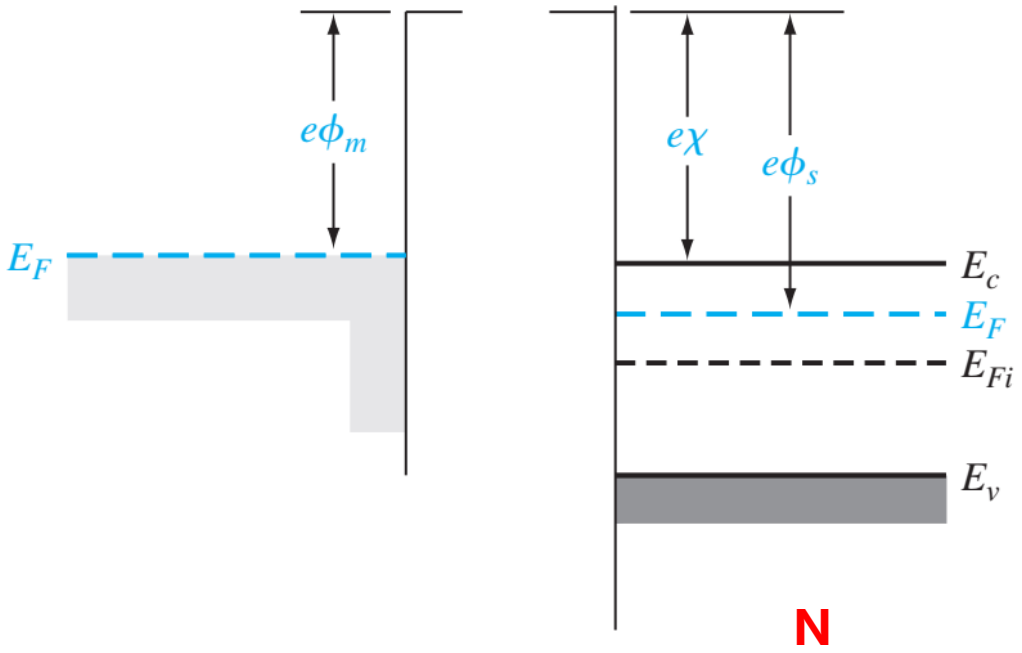
Two types of ohmic contacts:

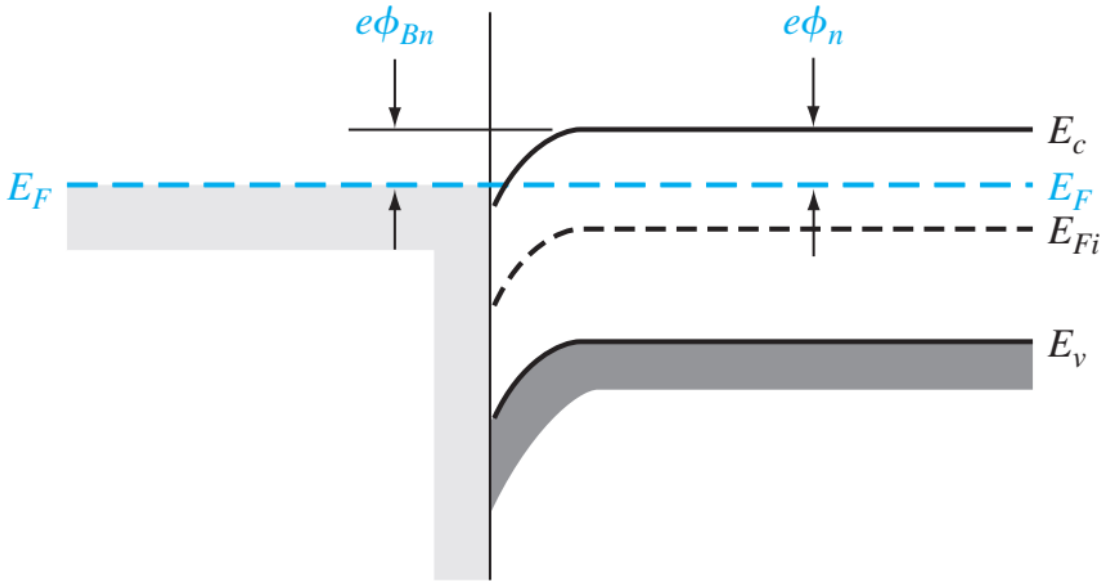
- Ideal nonrectifying barrier
- The tunneling barrier

9.2.1 Ideal Nonrectifying Barrier

Semiconductor: N Type, $\phi_m < \phi_s$

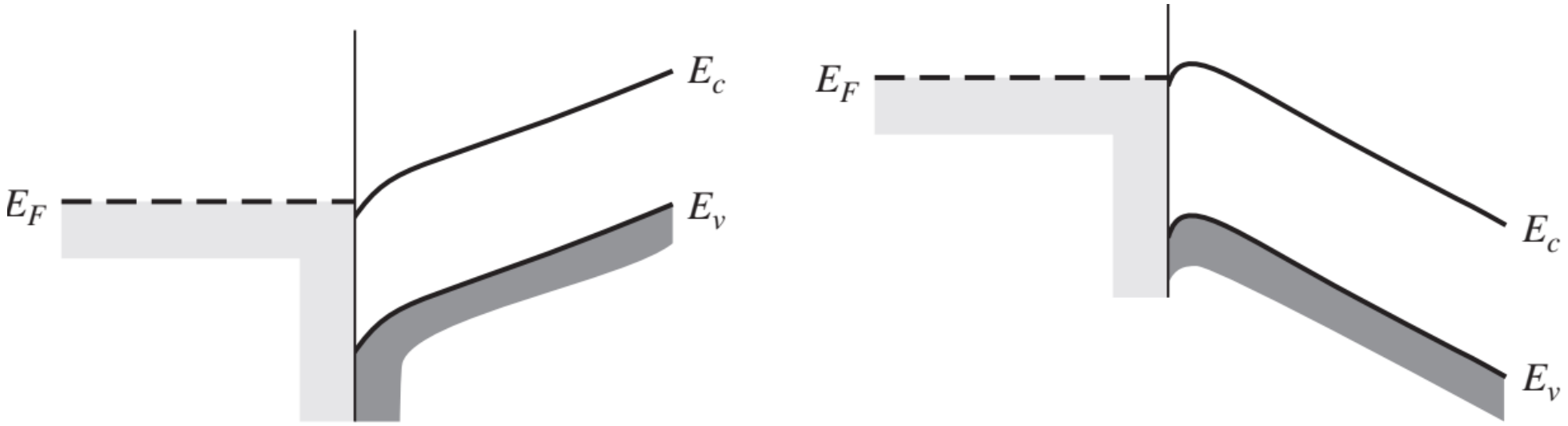
- The surface of the semiconductor <u>more n type</u>
- $\Phi_{Bn}=\Phi_n$, fairly <u>small</u> for a moderately to heavily doped semiconductor





9.2.1 Ideal Nonrectifying Barrier

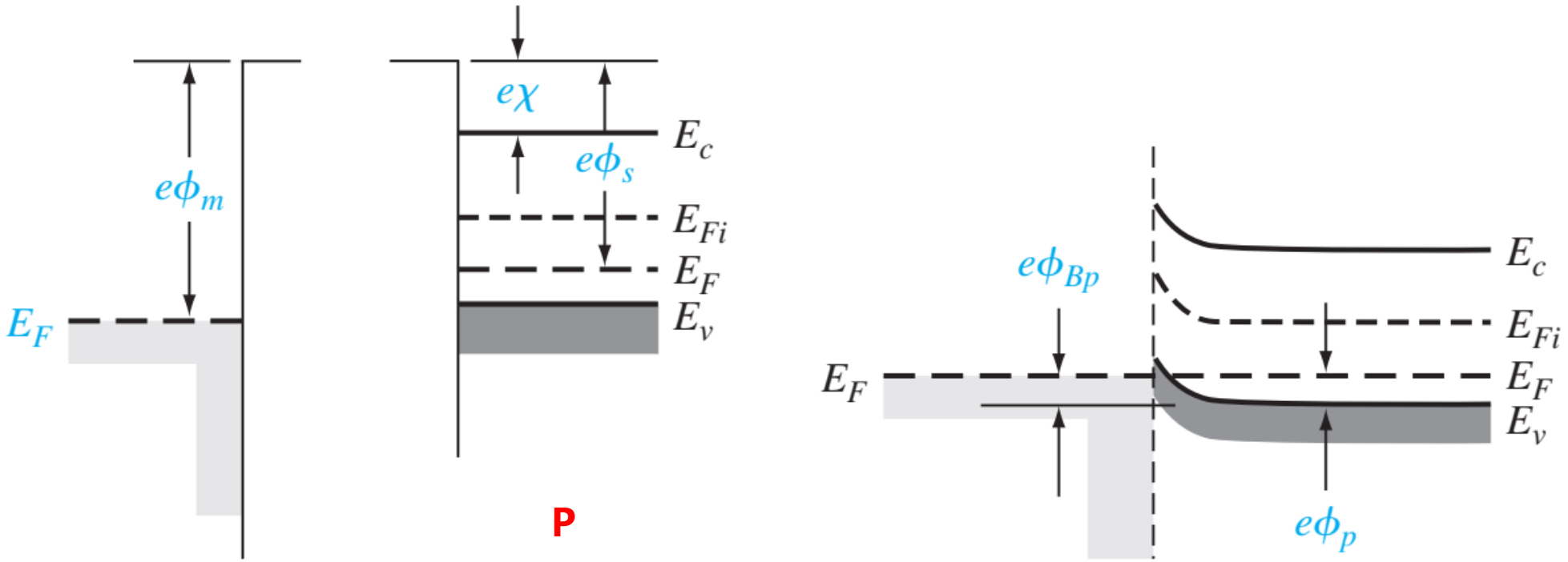
- If a positive voltage is applied to the metal, there is no barrier to electrons flowing from the semiconductor into the metal.
- If a positive voltage is applied to the semiconductor, the effective barrier height for electrons flowing from the metal into the semiconductor is fairly small.



9.2.1 Ideal Nonrectifying Barrier

Semiconductor: P Type, $\phi_m > \phi_s$

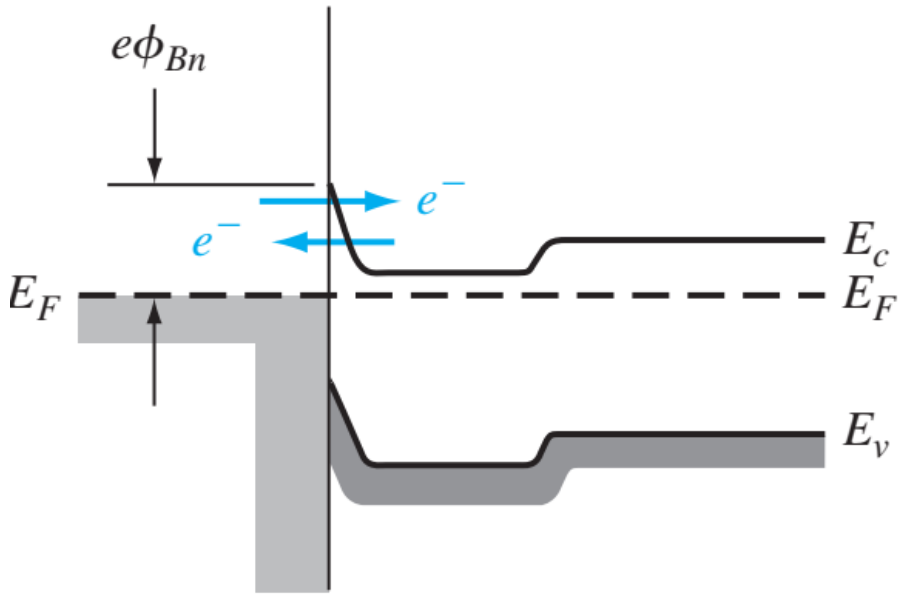
- The surface of the semiconductor <u>more p type</u>
- $\Phi_{Bp}=\Phi_p$, fairly <u>small</u> for a moderately to heavily doped semiconductor



9.2.2 Tunneling Barrier

Space charge region width

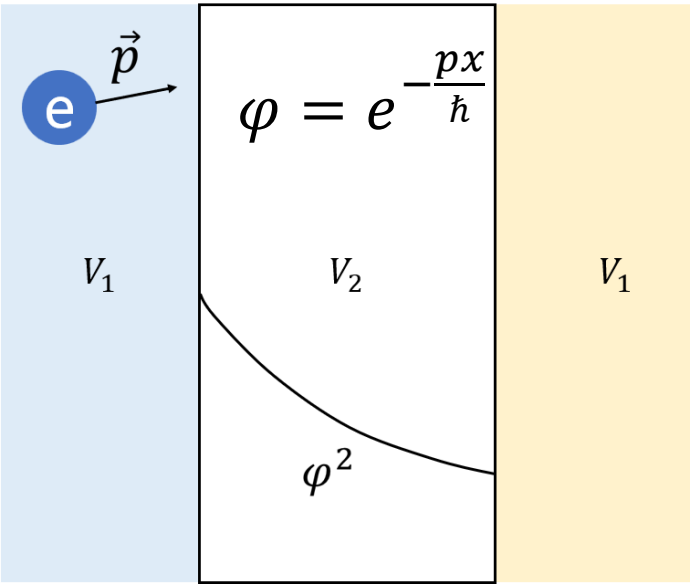
$$W = x_n = \left[\frac{2\epsilon_s(V_{bi} + V_R)}{eN_d}\right]^{1/2}$$



Tunneling current

$$J_t \propto \exp\left(\frac{-e\phi_{Bn}}{E_{oo}}\right)$$

$$E_{oo} = \frac{e\hbar}{2}\sqrt{\frac{N_d}{\epsilon_s m_n^*}}$$



9.2.3 Specific Contact Resistance

Contact resistance $R_c = \left(\frac{\partial J}{\partial V}\right)^{-1}\Big|_{V=0}$

Rc to be as small as possible for an ohmic contact

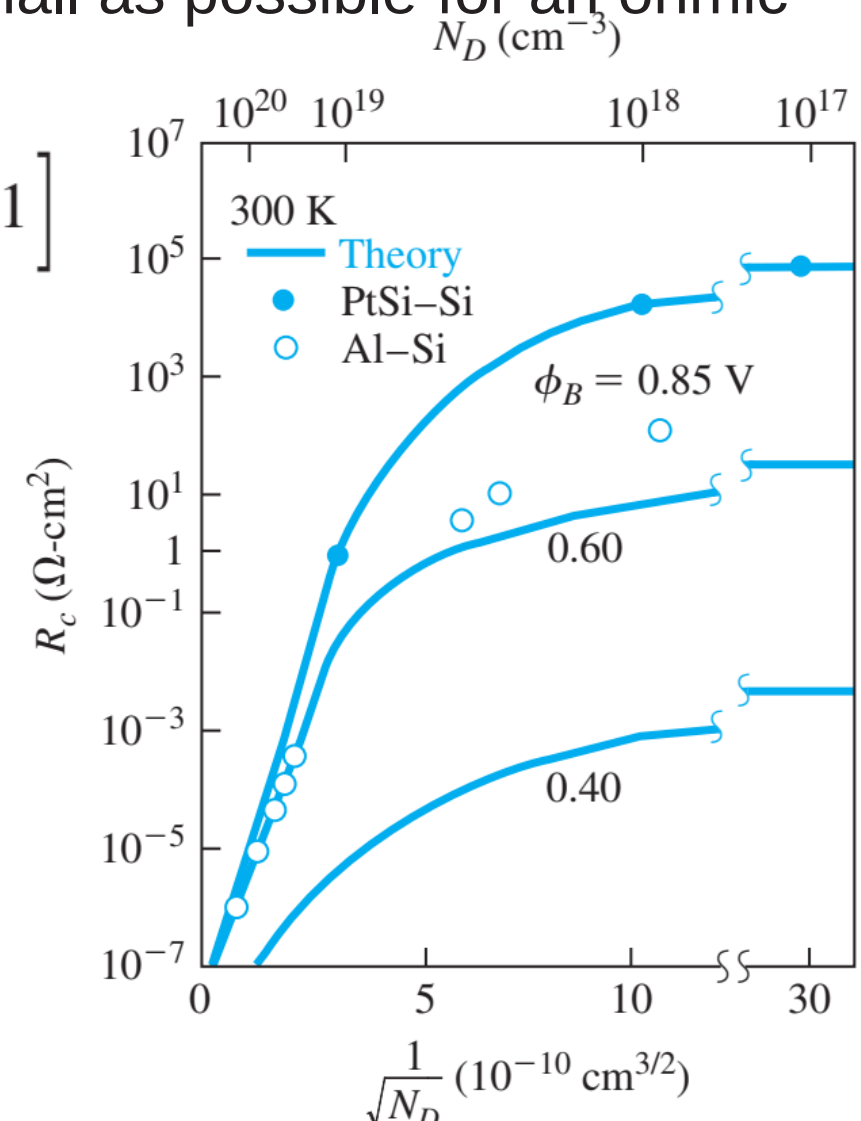
Rectifying contact $J_n = A^*T^2 \exp\left(\frac{-e\phi_{Bn}}{kT}\right)\left[\exp\left(\frac{eV}{kT}\right) - 1\right]$

(low to moderate doping
thermionic emission)

$$R_c = \frac{\left(\frac{kT}{e}\right)\exp\left(\frac{+e\phi_{Bn}}{kT}\right)}{A^*T^2}$$

(high impurity doping
tunneling)

$$R_c \propto \exp\left(\frac{+2\sqrt{\epsilon_s m_n^*}}{\hbar} \cdot \frac{\phi_{Bn}}{\sqrt{N_d}}\right)$$



9.3 | HETEROJUNCTIONS

Homojunction: The semiconductor material is homogeneous throughout the structure.

Heterojunction: The junction formed by the contact between two different semiconductor materials.

9.3.1 Heterojunction Materials

Heterojunction materials:

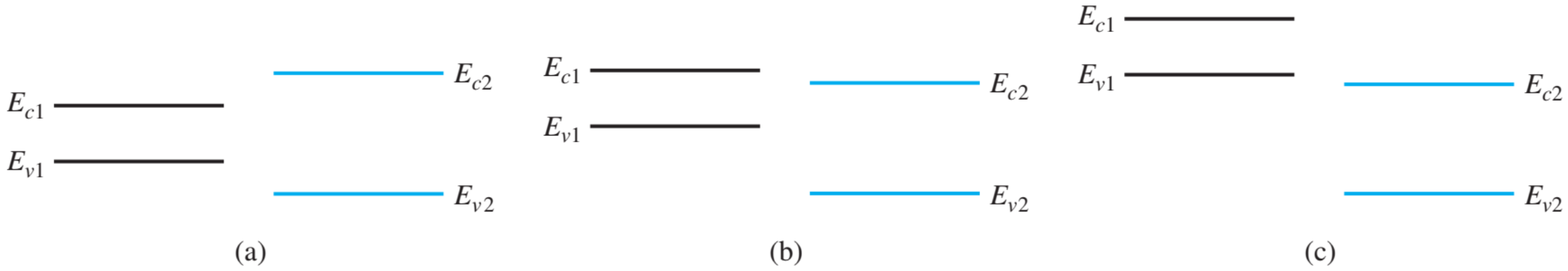
Different energy bandgaps discontinuity at the junction interface.

Useful heterojunction:

Lattice constant matching

Heterojunction	Lattice constant (10^{-10}m)	Lattice mismatch (%)
Ge-Si	5.6575~5.4307	4.1
Ge-InP	5.6575~5.8687	3.7
Ge-GaAs	5.6575~5.6531	0.08
Si-GaAs	5.4307~5.6531	4
Si-GaP	5.4307~5.4505	0.36

9.3.2 Energy-Band Diagrams

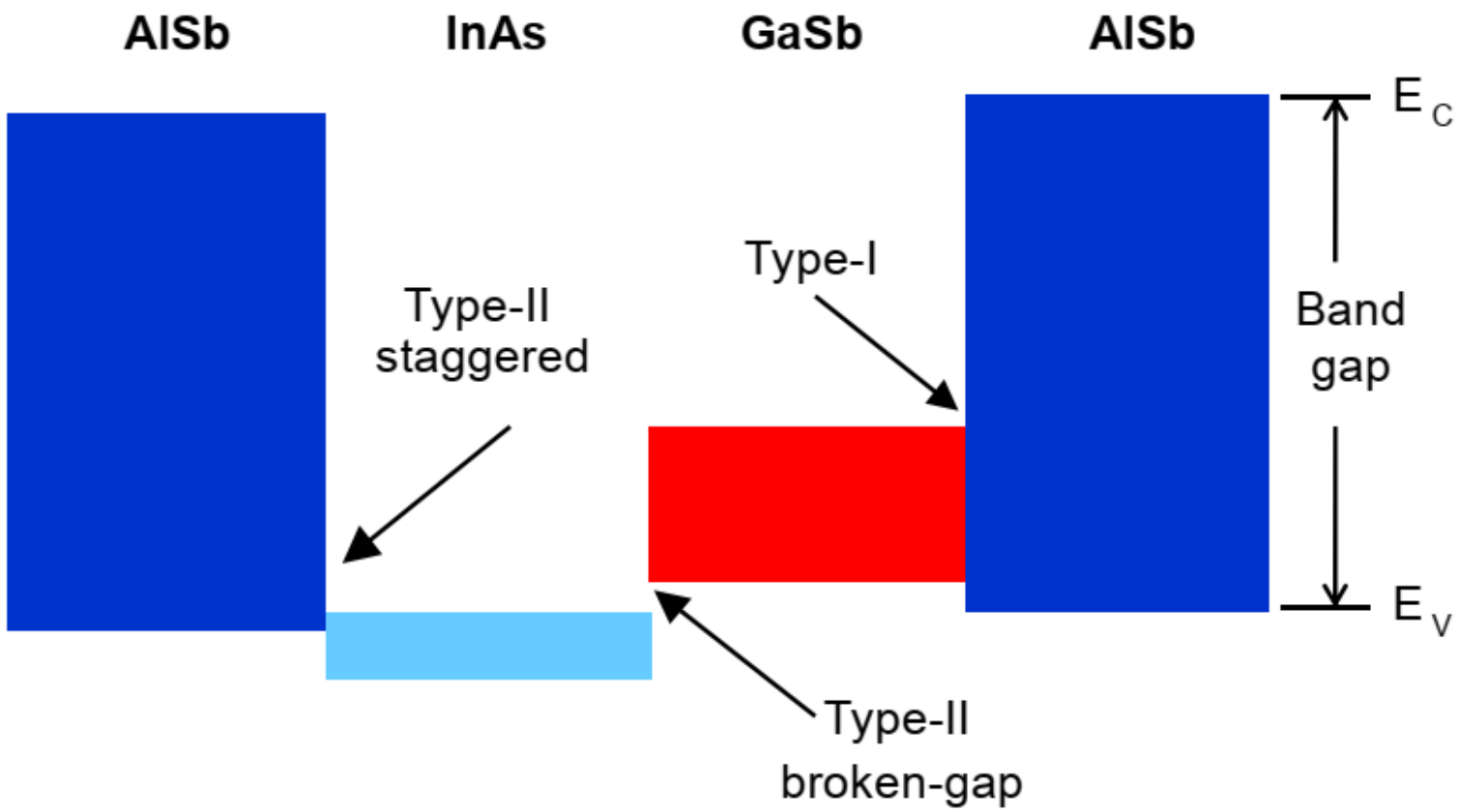


Straddling
跨骑

Staggered
交错

Broken gap
错层

9.3.2 Energy-Band Diagrams



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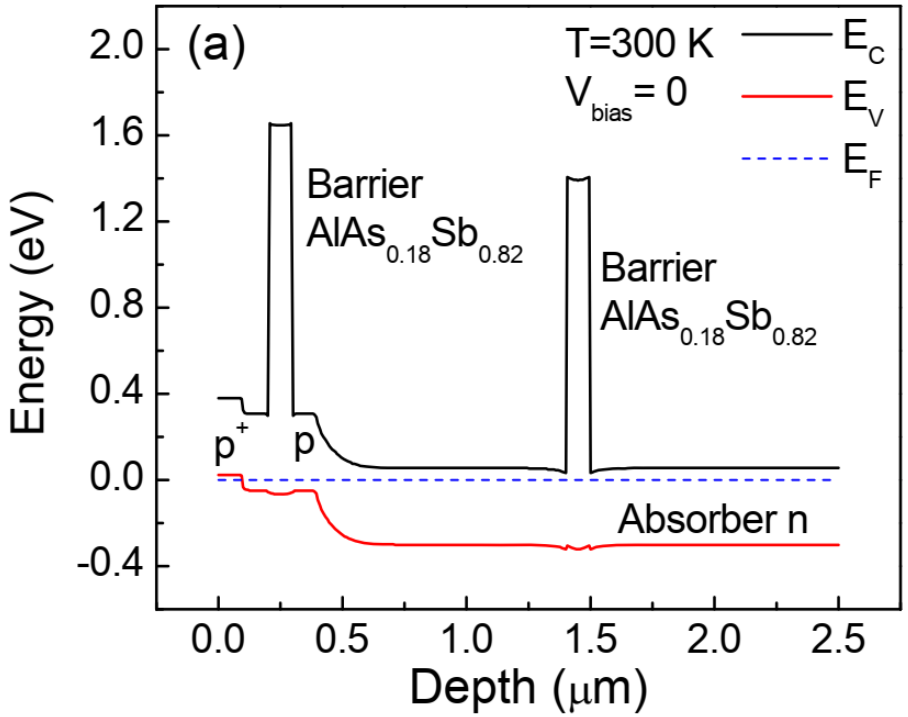

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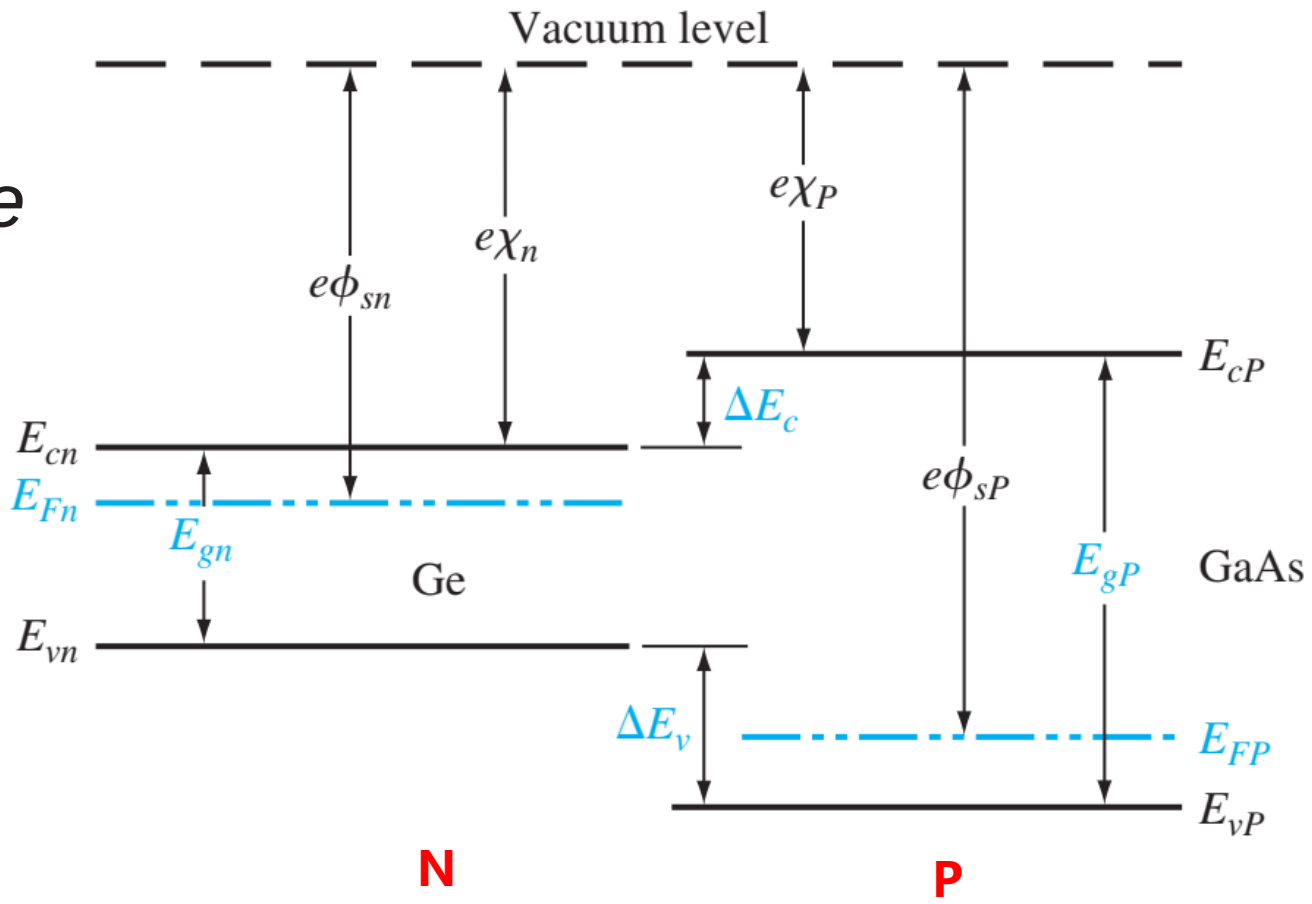
9.3.2 Energy-Band Diagrams

- **Before contact**

Ideal：electron affinity rule

$$\Delta E_c = e(\chi_n - \chi_P)$$

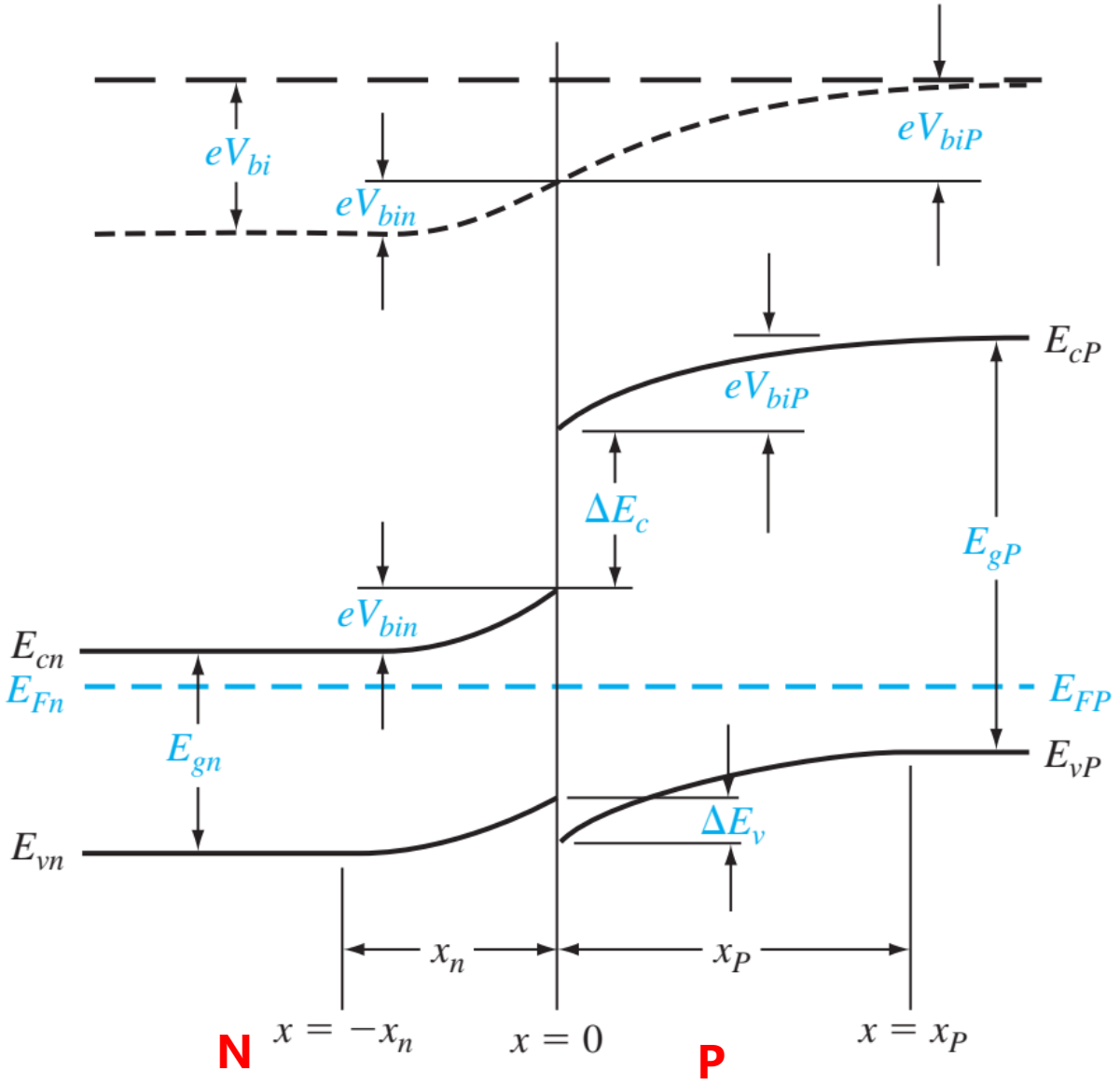
$$\Delta E_c + \Delta E_v = E_{gP} - E_{gn} = \Delta E_g$$



9.3.2 Energy-Band Diagrams

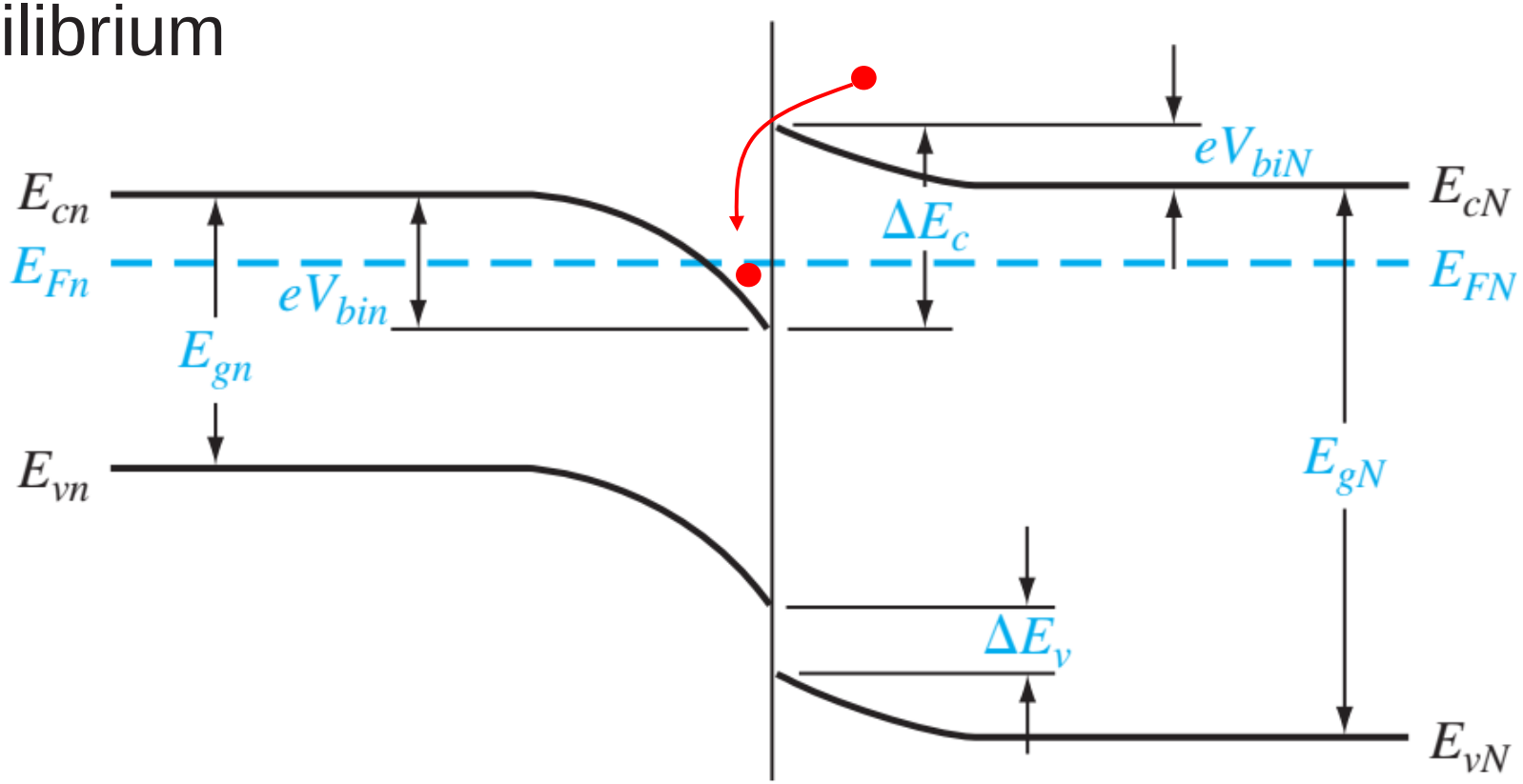
- **After contact**

electrons move from the n-side to the p-side, while holes move from the p-side to the n-side. The N-region forms a positive space charge region, while the P-region forms a negative space charge region. Additionally, there is a bending of energy bands.



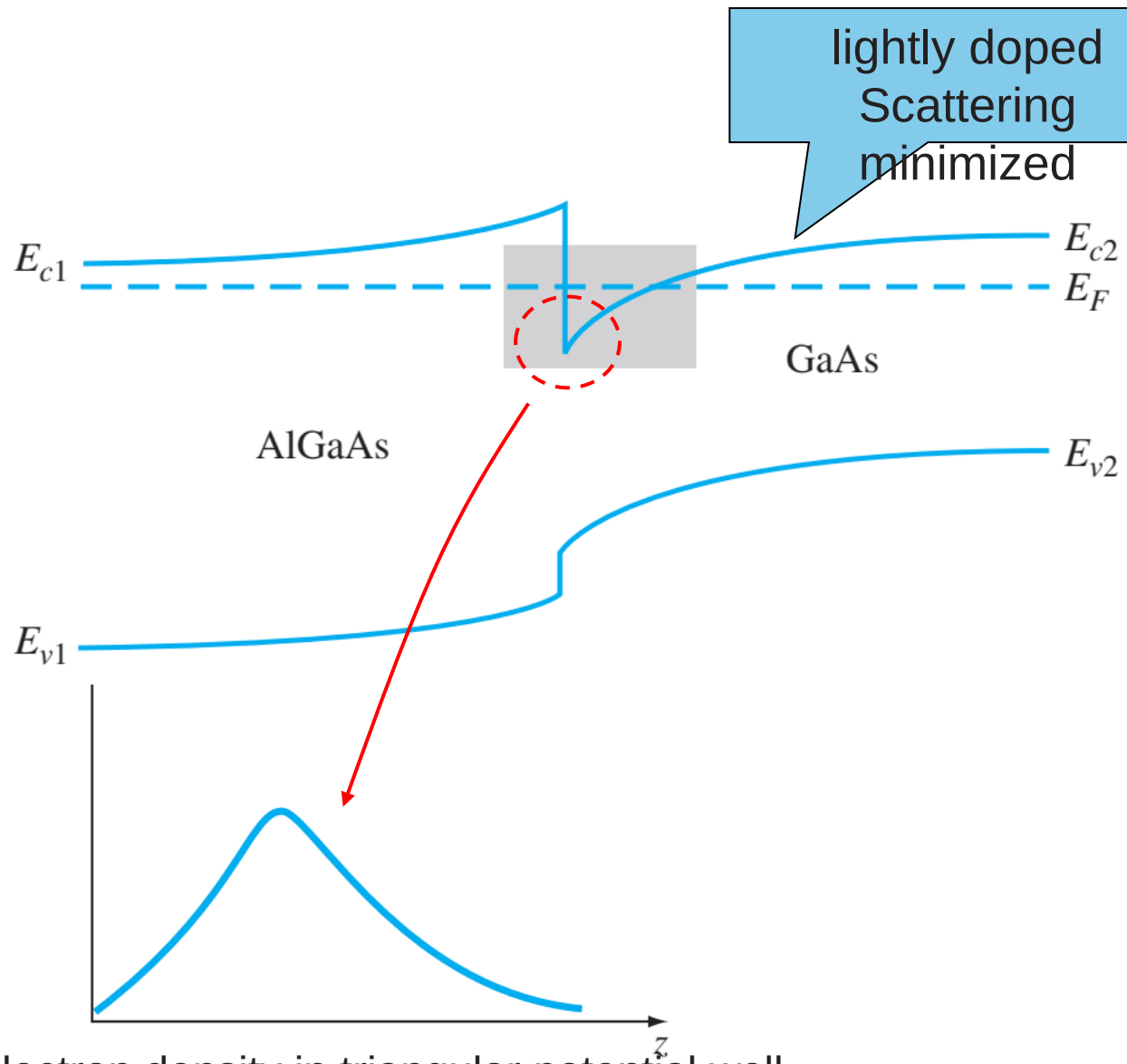
9.3.3 Two-Dimensional Electron Gas

- Ideal energy-band diagram of an nN heterojunction in thermal equilibrium

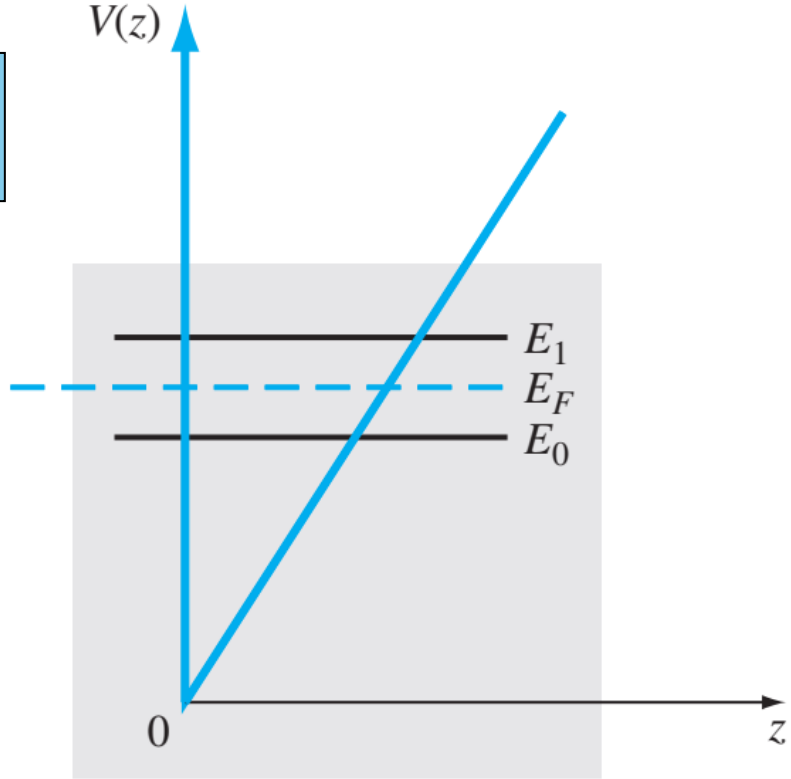


n GaAs (lightly doped or even intrinsic) –N AlGaAs (moderately to heavily doped)

9.3.3 Two-Dimensional Electron Gas

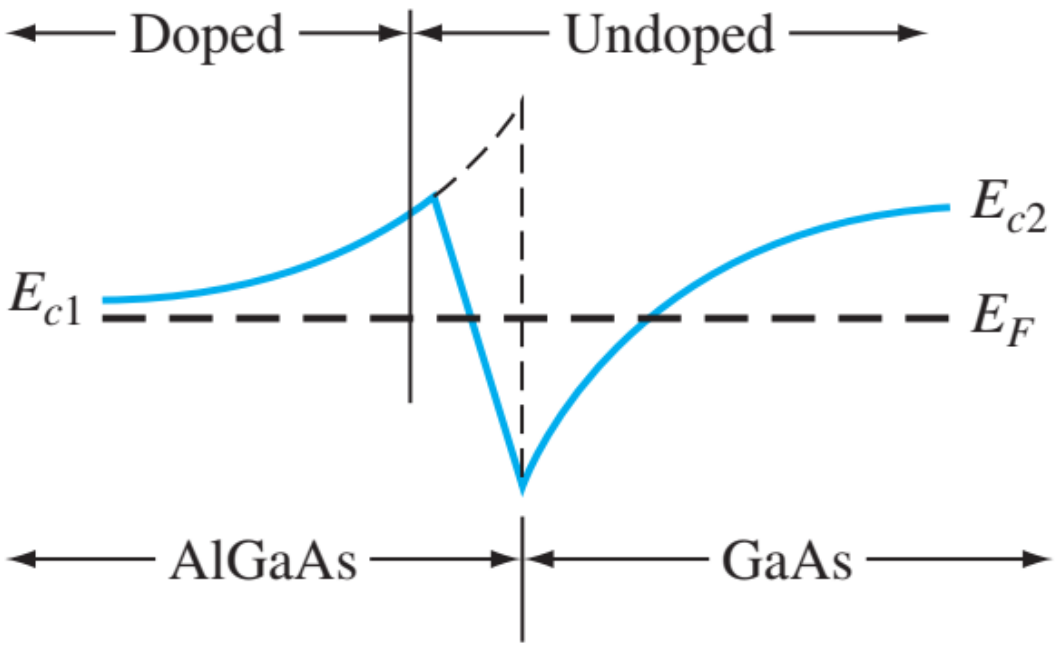


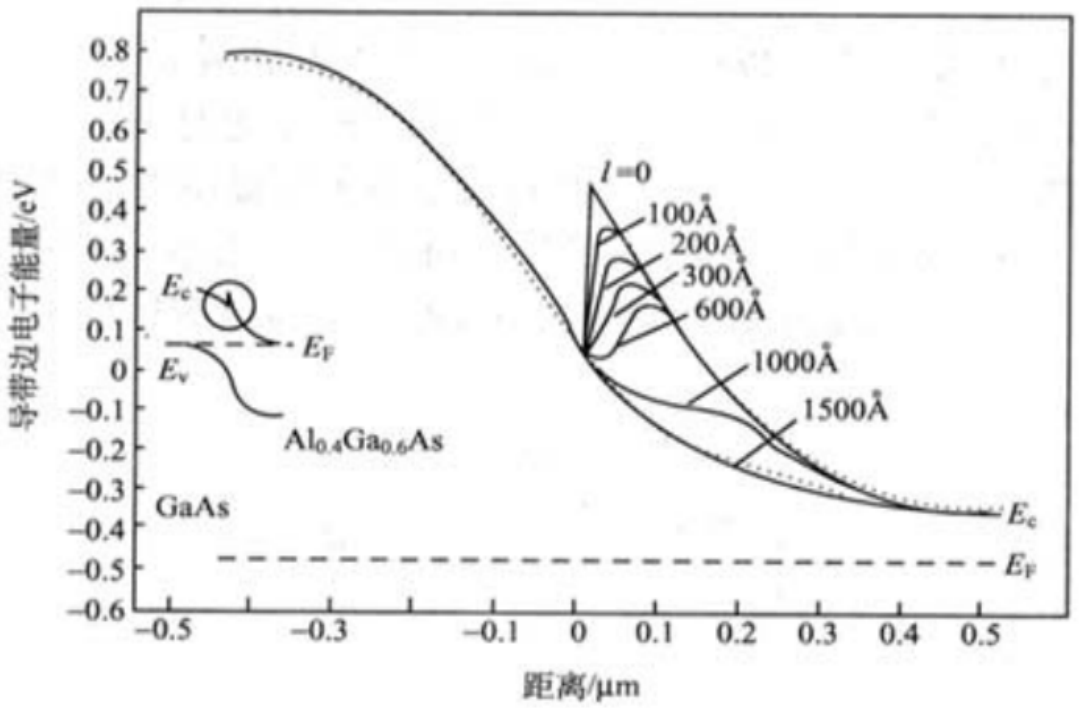
Electron density in triangular potential well.



triangular well approximation
with discrete electron energies

9.3.3 Two-Dimensional Electron Gas





Conduction-band edge at a graded heterojunction.